

Chapter 173

Recording

It's possible to record to one or more tracks on the Fairlight page, accommodating workflows as varied as editors recording scratch voiceover or temp sound effects, recording engineers recording narration, ADR, or foley as part of the audio finishing process, music studios recording orchestras for the music score, or garage bands recording their latest magnum opus.

While DaVinci Resolve is a comprehensive post-production environment for cinema and video, the Fairlight page can be used for any audio recording application you might have, from books on tape to live music to feature films and television.

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Setting Up to Record

Depending on how your workstation is set up, it's possible to simultaneously record to multiple tracks in the Fairlight page at once. How many tracks you can record to depend entirely on what hardware you have available. This section describes the process of recording to tracks in the Fairlight page.

Patching Inputs

Before recording audio anything, you need to use the Patch Input/Output window to patch an available audio input to a track.

DaVinci Resolve allows you to designate a full-time default input that is automatically patched to the track input when the R (Record Arm) button is clicked or manually patching in an audio input before record-arming it.

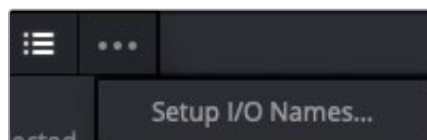
If you don't have an audio interface connected, you can use your workstation's inputs to connect whatever audio device is set up as the default audio input on your system to the track you want to record on.

Automatic Input Patching

The steps in this section explain how you can configure DaVinci Resolve to automatically patch in a designated default audio input whenever you click the R (Record-Arm) button in the track header or mixer channel of a track you want to record on.

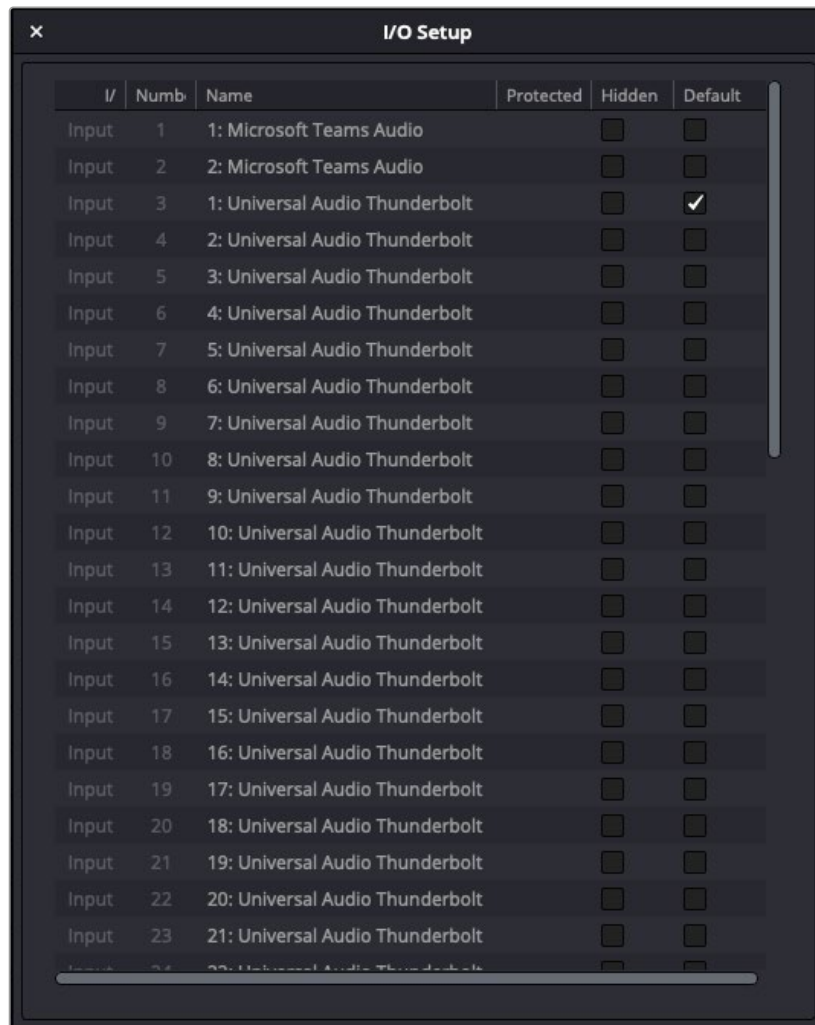
To set up Automatic Input Patching:

- 1 Click the Fairlight menu > Patch Input/Output.
- 2 When the Patch Input/Output panel opens, select Setup I/O Names in the Options menu in the upper-right corner. This opens the I/O Setup panel.



The Patch Input/Output Options menu

- 3 Place a checkmark in the Default column for the audio input you want to designate as the default track input source, then close the I/O Setup panel.



Default audio input selection in the I/O Setup panel

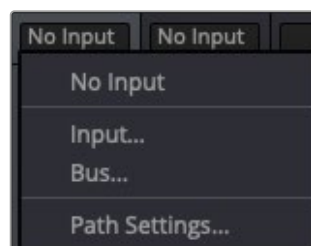
This input is automatically patched in when you record-arm any audio track using the R button in the Track header or Mixer channel. The audio input name will appear in the channel strip's input section.

Manual Input Patching

You can follow the steps below to manually select an input source without affecting the default audio input you may have already set up.

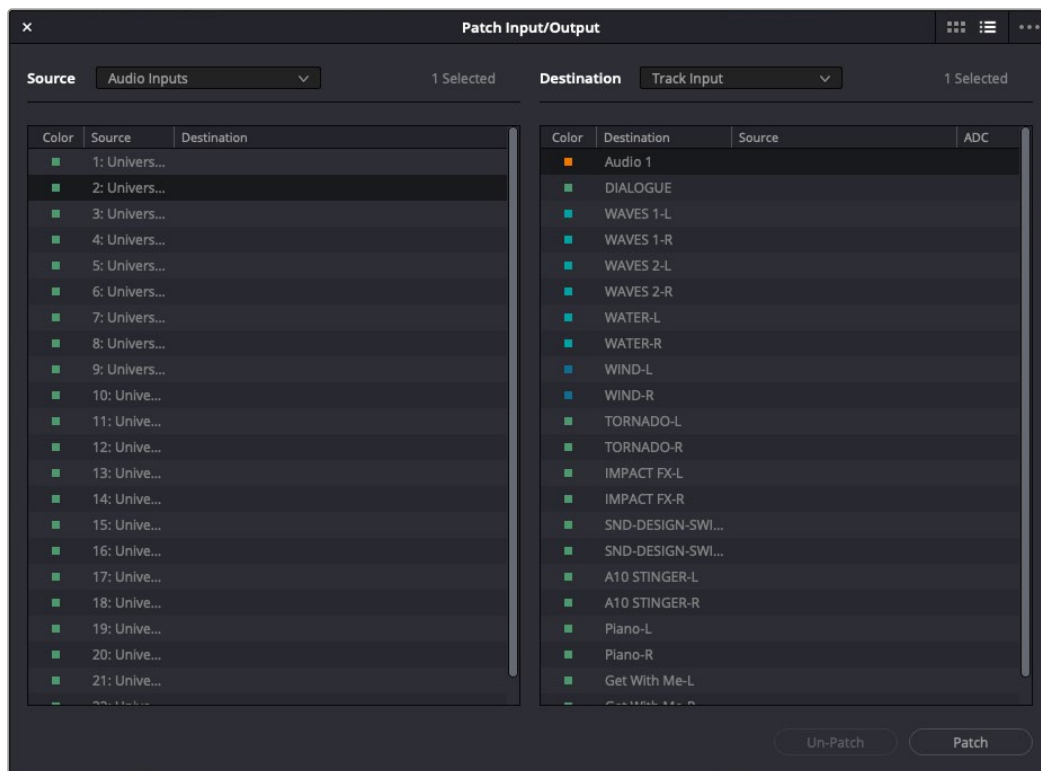
To manually patch in an auto input:

- 1 Open the Patch Input/Output pane by clicking the Input button at the top of the Mixer channel you want to record on and select Input. Alternatively, you can select Patch Input/Output from the Fairlight menu.



Mixer channel strip Input button menu

- 2 When the Patch Input/Output panel opens, select the audio input source and destination track input. You can repeat this step until you've patched all the inputs you want to record from to all the tracks you want to record to. You can patch as many inputs to as many tracks as your system can accommodate.
- 3 Click the Patch button in the lower-right corner of the panel.

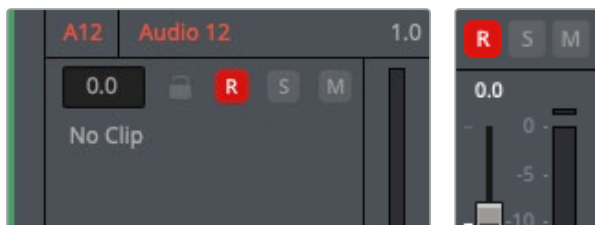


The Patch Input/Output panel with an Audio Input and Track Input selected

Arming Tracks

When you're ready to record on a track, click the R (Record Arm) button on either the Track header or corresponding Mixer channel.

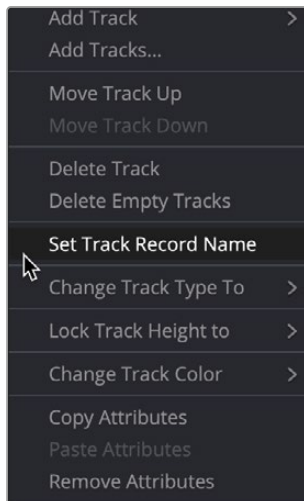
Audio tracks cannot be record-armed unless they've been patched using one of the methods explained above.



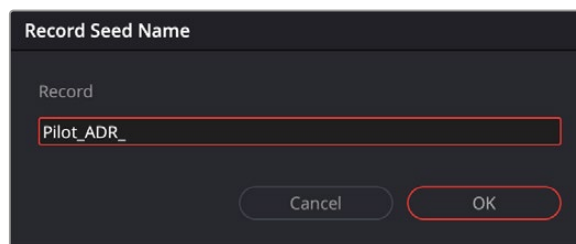
(Left) The Record Arm button activated on a Track header;
(Right) the Record Arm button activated on a channel strip

Record Name Prefix

A right-click on the designated record track header has the option to set a recording name prefix to the recordings for that track. This is a useful way to keep tabs of the various recordings required by your project. For instance, if recording ADR you can add a prefix for each character's recordings as in the example below for a character named Pilot.



Right-clicking a track header reveals the Set Track Record Name option



Example of a recording name prefix for a character called Pilot

Choosing Where to Record Audio Clips To

The process of recording in the Fairlight page creates new clips and generates additional media on disk. You can specify the location on disk where you want to save these recordings by opening the Capture and Playback panel of the Project Settings. In the Capture section use the Browse button, found underneath the "Save clips to" field, to choose a new location (a folder named "Capture" on your scratch disk is the default location).

To choose where the new clips that are created are placed in the Media Pool, simply open the Media Pool and select any bin in the Bin list, or create a new bin and select it if you want to put your recordings in their own location.

User-Selectable Input Monitoring Options

The Fairlight > Input Monitor Style submenu presents five options governing how you want to monitor inputs while recording.

- **Input:** You only hear the live signal being input; you never hear the contents of tracks.
- **Auto:** When one or more tracks are armed for recording, you hear the live input signal; on playback you hear the contents of each track.
- **Record:** You only hear the live input signal while actively recording, meaning the Record button has been pressed while one or more tracks are armed for recording. You don't hear the input signal while tracks are merely armed.
- **Mute:** You hear nothing.
- **Repro:** While recording, you only hear what's just been recorded, played from the track. In other words, you're not listening to the live input, but you're reviewing what's just been recorded as it's recording.

Recording Using the Onscreen Controls

You can record anywhere you want on the currently armed track or tracks by placing the playhead where you want recording to begin. In this way, you can record to specific areas of your program as you record voiceover, sound effects, foley, or other timed performances that need to fit into a particular region of the edit.

To begin recording:

- 1 Position the playhead where you want recording to begin.
- 2 Click the Record button in the transport controls. Recording immediately begins, and the material being recorded immediately begins drawing a waveform in real time, giving you immediate feedback that the input you're recording is properly connected or not, as well as where on the currently armed track material is being recorded.

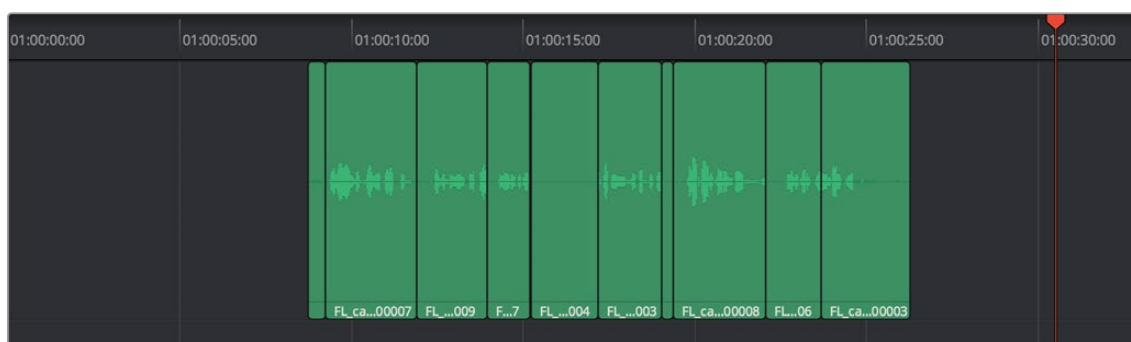
To stop recording, do one of the following:

- Click the Stop button in the transport controls.
- Press the Spacebar.

Recording and Editing Multiple Takes Using Layering

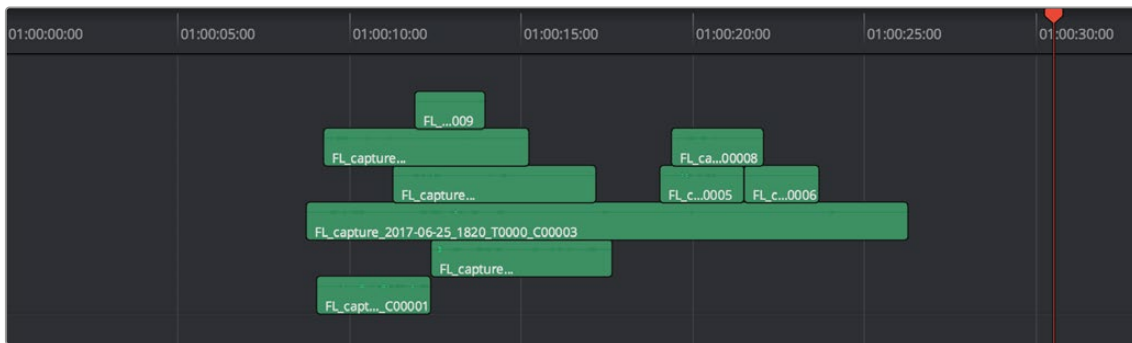
There are two ways you can record multiple takes. You can either record them one after the other, sequentially, and then edit them later. However, you also have the ability to record multiple takes to the same region of the timeline, one on top of another, while at the same time preserving every take using track layering.

In the following screenshot, multiple takes have been recorded over the same section of the timeline, including some partial takes to correct specific phrases in the voiceover being recorded. When you do this, the result looks like a series of cuts and overwritten clips, with the most recently recorded segments being the ones that play back over the previously recorded segments.



Overlapping recordings with Audio Track Layers turned off

However, if you choose View > Audio Track Layers, you'll see that all your recordings have actually been preserved via a vertical stack of overlapping audio clips.



Overlapping recordings with Audio Track Layers turned on, showing layering within the same track

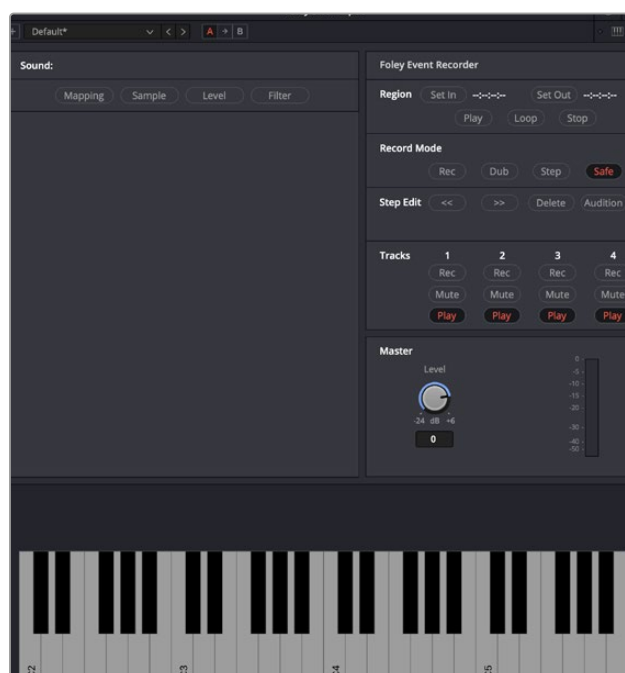
The layering of audio clips in DaVinci Resolve means that the topmost superimposed clips in a layered stack like this mutes the audio of overlapping clips that are lower in the stack.

Using layering, it's easy to edit the best segments of the best takes, while preserving all other takes, simply by adding edits and rearranging clips in the stack so the best parts are on top.

For more information about audio layering, see *Chapter 175, “Editing Basics in the Fairlight Page.”*

Recording VSTi Instruments

DaVinci Resolve supports VSTi instruments working with connected MIDI controllers to trigger instrument sounds that can be recorded live on audio tracks of the Timeline. DaVinci Resolve Studio includes the Fairlight FX Foley Sampler. This can be loaded with foley sounds such as footsteps or human movements, so you can perform these sounds in real time and record the result to another track as you watch performers walking or punching in the edit, even if you lack a recording booth with foley pits and props.



The Foley Sampler ready to be filled with sounds

For more information about the Foley Sampler, see *Chapter 181, “Fairlight FX.”*

On the other hand, if you’re a musician, there’s nothing stopping you from loading VSTi musical instruments of different kinds for playback, and using the Fairlight page as a multi-track recorder. DaVinci Resolve doesn’t have MIDI sequencing functionality, but you can record live playback straight to the Timeline, using layered audio to manage multiple takes for later re-editing. Bet you never thought you’d be recording music in DaVinci Resolve.



A VST Instrument (in this case Serato Sample) loaded into a track of the Timeline

To enable a MIDI controller in macOS:

- 1 If DaVinci Resolve is running, quit before connecting your MIDI controller and setting it up.
- 2 On macOS you'll use the Audio MIDI Setup utility to choose output hardware and select a speaker configuration to be made available on your system. In the Finder, use Spotlight and search for Audio MIDI Setup to open it.
- 3 In Audio MIDI Setup, choose Window > Show MIDI Studio. A window showing icons for all connected MIDI controllers appears. Your controller should be showing an icon. If it's not, you may need to install drivers for it.
- 4 Select the icon for your controller and turn on the "Enter test MIDI setup mode" button (it looks like a little keyboard) to test if your keyboard is connecting with the computer. If it is, then turn this off.

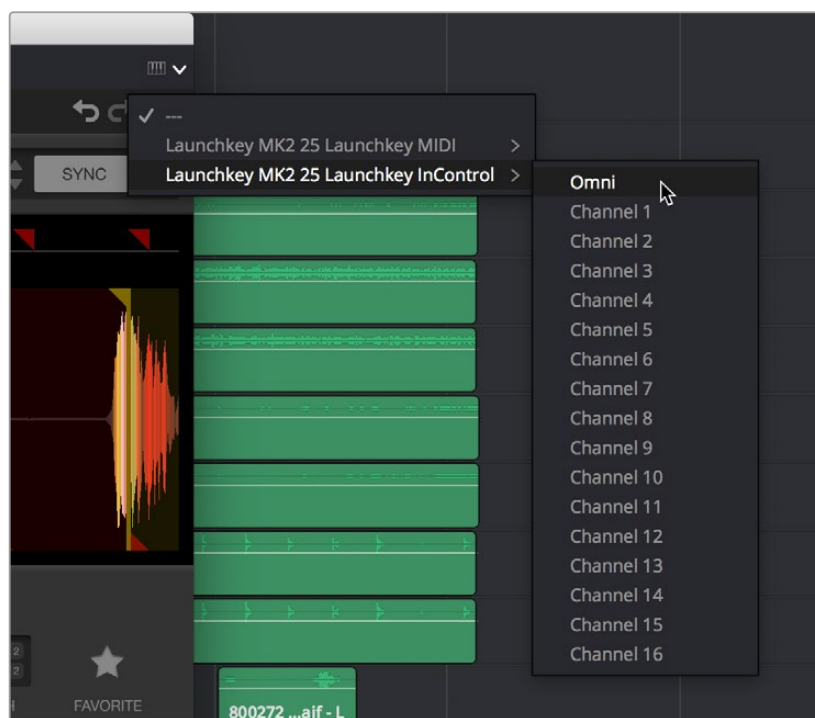
For more information on setting up MIDI on different systems, see the DaVinci Resolve Configuration Guide, available on the web from the Blackmagic Design support page at www.blackmagicdesign.com/support/family/davinci-resolve-and-fusion.

To set up the Fairlight page for VSTi instrument recording using a sampler:

- 1 Open DaVinci Resolve.
- 2 Make sure you have at least two available audio tracks in the Timeline, one for the instrument you'll be playing, and one to record into. This example will use tracks A4 and A5 for this.
- 3 Open the Effects Library, find a VSTi sampler you have installed on your system, and drag it to the track header of the track you want to use for playing, for example track A4.

Massively-featured sampler/synth combinations such as Native Instruments Kontakt and Steinberg Halion are ubiquitous and useful when you want to specifically map a collection of sound effects to specific keys or pads to create re-usable multi-purpose instruments. However, more streamlined samplers that emphasize automatic audio clip slicing such as Serato Sample (Windows and macOS) or Image Line Slicex (Windows only) can make short work of the more specialized task of loading library sound effects recordings (or custom recordings you create) with multiple footsteps, punches, keyboard presses, cloth rustles, or other foley activities, and quickly splitting them up into individually playable samples you can trigger with pads or a keyboard.

- 4 When the VSTi interface window appears, open the MIDI menu at the upper right-hand corner of the VSTi window and choose the correct MIDI channel from your MIDI controller's submenu. If you've selected the correct MIDI channel, the instrument should start responding to the keys or pads on your controller.



Enabling MIDI control

- 5 Next, configure the VSTi instrument you're using to play the sound effects you want to use for foley. In this example, the Serato Sample VSTi plugin is being used to automatically slice up a recording of footsteps from one of Sound Ideas' many sound effects libraries.

Because the VSTi you added is patched to that track's Insert (if you look at the Mixer you should see that the I button is enabled on the channel strip the instrument is patched to), the Send is PRE the Instrument. This means you need to patch that track's Track Direct output to the input of another track to record the instrument.

- 6 Choose Fairlight > Patch Input/Output to open the Patch Input/Output window, then set the Source dropdown menu to Track Direct and the Destination dropdown menu to Track Input. Click Audio 4 to the left, and Audio 5 to the right, and click the Patch button; this sets you up to play the VSTi plugin on track A4, and record its output on track A5.

Be aware that after patching Track Direct from the track with the instrument to the track you're recording onto, you also need to turn "Direct Output" on for that track in the Path Settings of that track's channel strip in the Mixer.

- 7 Open the Mixer (if necessary), click the Input dropdown menu at the top of the channel strip that shows the VSTi instrument you're using, and choose Path Settings. When the Path Settings window appears, click the ON button for Direct Output, then close the Path Settings window.

At this point, you're ready to begin recording.

To play and record a VSTi instrument:

- 1 Click the Record Arming button of the track you're recording to (in this example A5), move the playhead to where you want to begin recording, and then click the Record button to begin recording.
- 2 As the video of your program plays, use your MIDI controller to trigger sound effects as necessary. When you're finished, click the Stop button.

If necessary, you can record multiple takes using track layering until you get the timing right. When you're finished, you can remove the instrument from the track it's on since the recorded audio is all you need.